

Experiment 8

Newton's Second Law

Purpose of the Investigation

To investigate the relationship between the variables force, mass and acceleration.

Background information

Newton's Second Law states that the rate of change of momentum is directly proportional to the unbalanced force acting on the body and is in the direction of the force.

$$F \propto \frac{mv - mu}{t}$$

The S.I. system of units is such that the constant of proportionality is 1, which gives:

$$F = \frac{mv - mu}{t}$$

If m is constant then

$$F = m \frac{(v - u)}{t}$$

which gives

$$F = ma$$

which is an alternative form of Newton's Second Law.

In most practical examples of motion, the force of friction is always acting. This force, which opposes motion, must be balanced for a body to maintain constant velocity. Therefore to achieve a positive acceleration, an object must have a force greater than the frictional force acting on it.

Thus when a body has constant velocity, any force then added will cause an acceleration of the body in the direction of the force.

Equipment required

- Aluminium foil
- One stop clock (Unilab or similar)
- Six electrical wires
- Balance
- Pan for masses
- String
- Dynamics trolley with metal wheels
- Set of small slotted masses (10 x 50 g)
- Large mass (0.5 kg mass)
- Pulley
- Clamp
- One switch

Part I - Variable force and constant mass

Procedure

#1 Set up the equipment as shown in the Figure 8.1 and 8.2.

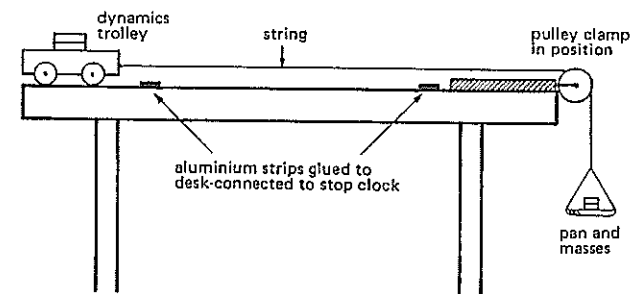
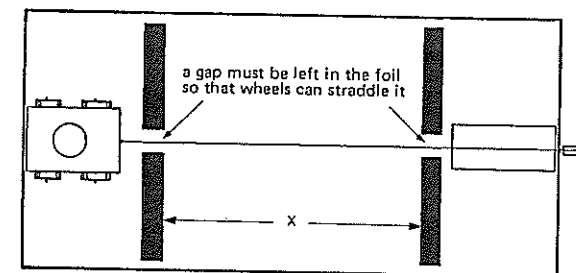


Fig 8.1 Elevation



x must be less than the distance the pan falls

Fig 8.2 Plan view

#2 Connect a stop clock to the aluminium foil as shown in Figure 8.3. This measures the time taken for the trolley to go from AB to CD.

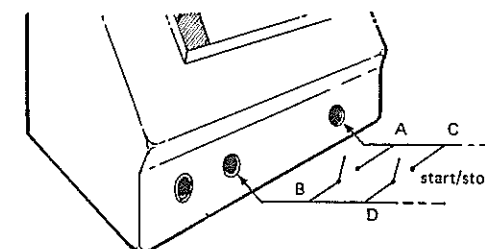


Fig 8.3

#3 Place ten 50 g slotted weights onto the trolley.

- #4 To overcome the effect of friction it is necessary to place masses in the pan. These act as a counterweight which can balance the frictional forces. The system is balanced when the trolley moves with constant velocity after it is given a slight push. Any additional mass in the pan will provide the accelerating force. You are now ready to record the additional force and determine the acceleration.
- #5 In each case you will need to calculate the accelerating force in newtons acting on the pan. Remember, gravity exerts a force on all masses.
- #6 While holding the trolley, take one 50 g mass from the trolley, and place it into the pan. This ensures that the total mass accelerated by the increasing force remains the same throughout the experiment.
- #7 Place the leading wheels of the trolley approximately 1cm in front of AB. Ensure that the wheels will straddle the gap in AB and CD.
- #8 After resetting the stopclock, release the trolley and record the time and accelerating force obtained in a table similar to the one below.

Trial	Accelerating force (N)	Time (s)	Acceleration (m s^{-2})
Force 1			
Force 2			

- #9 Take another 50 g mass from the trolley and place it with the other in the pan.
- #10 Again reset the timer and release the trolley.
- #11 Repeat the above steps adding more 50 g masses to the pan from the trolley to give readings for Forces 3 to 6.
- #12 Measure the mass of the trolley, the pan, the 500 g and the counterweight, and record the total mass below the table.
- #13 Measure the distance between AB and CD and also record this value below the table.

Processing of the results

- 1 By using $s = ut + \frac{1}{2}at^2$ with $u = 0$, calculate the acceleration for each run. Enter these values in your table.
- 2 Graph accelerating force on the vertical axis against acceleration on the horizontal axis.
- 3 Use your graph to determine a value for the gradient.

Conclusion

From the results you obtained in this equipment, describe the relationship between the accelerating force and the resulting acceleration when the mass of the system is kept constant.

Questions and discussion

- 1 What is the shape of the graph of accelerating force against acceleration?
- 2 What is the mathematical term to describe the relationship between the accelerating force and the acceleration?
- 3 Write your answer to Question 2 using mathematical symbols F and a only.
- 4 Write down the value you obtained for the gradient of your graph.
- 5 Can you relate the value of the gradient to any of the measured data from the experiment. What is the significance of this relationship?

Part 2 - Variable mass and constant force

Procedure

- #1 Set up the equipment as for Part 1 of this experiment but have no extra mass on the trolley.
- #2 Add masses to the pan so that the trolley will just move with constant velocity. This is the counterweight.
- #3 Hold the trolley and place an extra 100 g in the pan.
- #4 Record the mass of the trolley, pan, counterweight and 100 g in a table like the one below. This is the total accelerated mass.
- #5 Reset the timer.
- #6 Release the trolley.
- #7 Record the time for this trial.

Trial	Total accelerated mass (kg)	Time (s)	Acceleration (m s^{-2})	$\frac{1}{\text{Acceleration}}$
Mass 1				
Mass 2				

- #8 Reset the timer.
- #9 Release the trolley and record the time in your table.
- #10 Repeat the experiment with masses of 500 g, 700 g, 800 g, 900 g and 1000 g.
- #11 Measure the distance between AB and CD and record this value underneath your table.
- #12 Record the value of the constant accelerating force underneath your table.

Processing of the results

- 1 By using $s = ut + \frac{1}{2}at^2$, calculate the acceleration and the inverse of acceleration for each trial. Enter these values in your results table.

- 2 Graph accelerated mass on the vertical axis against acceleration.
- 3 Graph accelerated mass against the inverse of acceleration.
- 4 Calculate a value for the gradient of the graph produced in Question 3.

Conclusion

Describe from the results obtained in this experiment, the relationship between the accelerated mass and the acceleration when the accelerating force is kept constant.

Questions and discussion

- 1 What is the shape of the graph of
 - a accelerated mass vs acceleration, and
 - b accelerated mass vs the inverse of acceleration?
- 2 Write a sentence which describes the relationship between the accelerated mass and acceleration?
- 3 Write your answer to Question 2 using the mathematical symbols.
- 4 Write down the value you obtained for the gradient of accelerated mass and the inverse of acceleration.
- 5 Can you relate the value of the gradient to any of the measured data from the experiment?

Experiment 9

Conservation of momentum: a descriptive investigation

Purpose of the investigation

To investigate in a semi-quantitative manner the relationship between momentum before and momentum after in explosion and collision situations.

Background information

Momentum is conserved in any closed system on which no external unbalanced force is acting. Thus when two stationary dynamics trolleys are 'exploded' apart from rest, the sum of their momenta should remain zero since all the forces acting on the system are internal.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

but $m_1 u_1 + m_2 u_2 = 0$

therefore $m_1 v_1 + m_2 v_2 = 0$

In Part 1 of this experiment you will examine the law of conservation of momentum when two trolleys are 'exploded' apart.

In Part 2 of this experiment you will examine the conservation of momentum in a simple inelastic collision where a moving object collides with a stationary object.

Equipment required

Two dynamics trolleys (with spring mechanism)
 Four sets of weights (500 g)
 Two stop clocks
 Two metre rules
 Piece of chalk

Part 1: Examination of momentum changes in an 'explosion' collision**Procedure**

- #1 This experiment is best done on the floor.
- #2 Use a piece of chalk to mark a spot on the floor from where the trolleys will 'explode'.
- #3 On trolley 1 place three 500 g masses but do not place a load on trolley 2. Record your data in a table similar to the one outlined below.

Trial	Mass 1 (kg)	Mass 2 (kg)	Time 1 (s)	Time 2 (s)	Velocity 1 (m s ⁻¹)	Velocity 2 (m s ⁻¹)	Momentum (N s)	Momentum (N s)
1								

- #4 From the chalk mark on the floor measure a distance of 1m in one direction and a distance of 1m in the opposite direction. At these positions chalk in a line such that it is radially perpendicular to the initial chalk mark as shown in Figure 9.1.